

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T51 · Vectors

Questions

Student book · questions with writing space

■ 23 pages

FAMILY MAP

Vectors

F01 Use vectors with ratios and intersections

14 Q | 67 marks

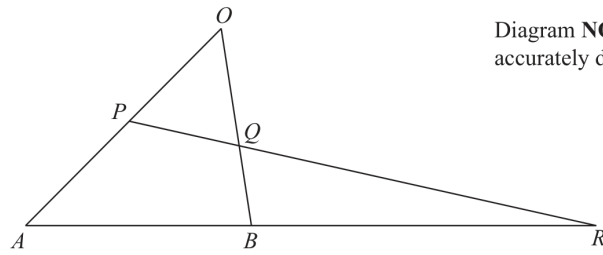
F02 Calculate with vectors

2 Q | 5 marks

F03 Read and represent vectors

1 Q | 1 marks

23

Diagram **NOT**
accurately drawn

OAB is a triangle.
 P is the midpoint of OA
 Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

.....
 (1)

Continued
4MA1-1H Q23

UNIT 2

23

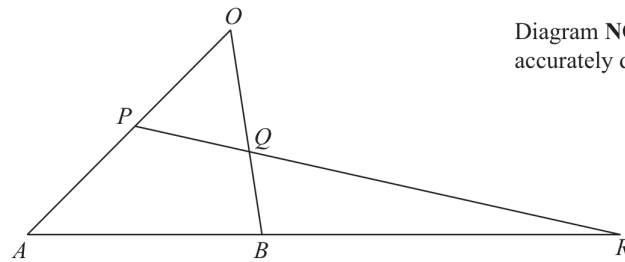


Diagram **NOT**
accurately drawn

OAB is a triangle.
 P is the midpoint of OA
 Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \dots\dots\dots (4)$$

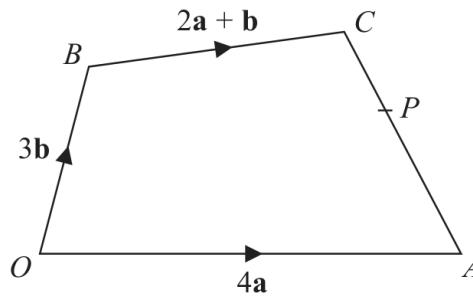
Question 26

4MA1-2H Q24

MODULAR UNIT 2

6 marks

24

Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

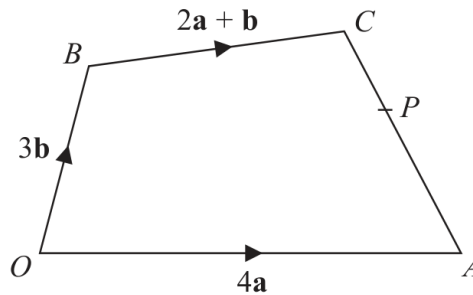
Question 26 (continued)

4MA1-2H Q24

MODULAR UNIT 2

6 marks

24

Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

Question 35

4MA1-2H Q22 (a)(i), (a)(ii), (b)

MODULAR UNIT 2

5 marks

22

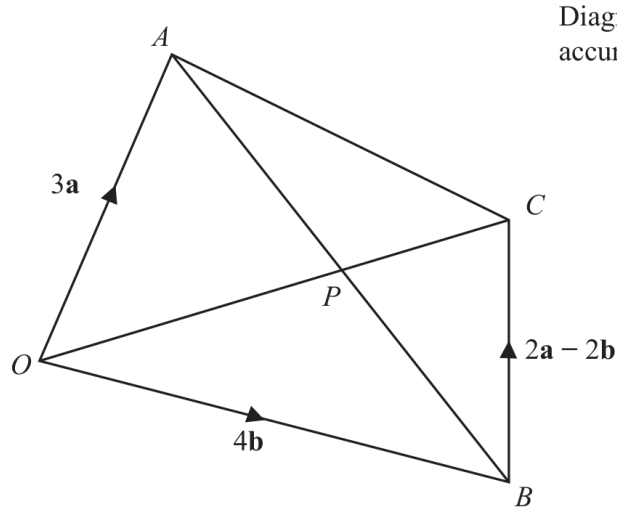


Diagram **NOT**
accurately drawn

$OACB$ is a quadrilateral.

$$\overrightarrow{OA} = 3\mathbf{a} \quad \overrightarrow{OB} = 4\mathbf{b} \quad \overrightarrow{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\overrightarrow{OC}$ in terms of **a** and **b**.
Simplify your answer.

$$\vec{OC} = \dots\dots\dots (1)$$

- (ii) Find the vector $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\overrightarrow{AB} = \dots\dots\dots (1)$$

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

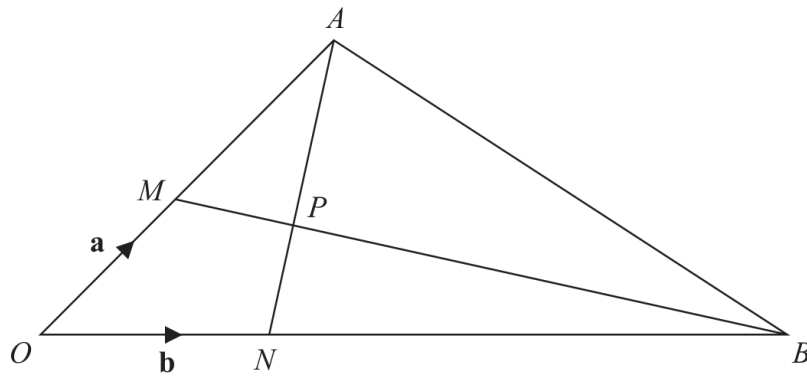
Question 15

4MA1-1H Q19

MODULAR UNIT 2

5 marks

19

Diagram **NOT**
accurately drawn OMA, ONB, MPB and NPA are straight lines. M is the midpoint of OA $ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vector $\vec{AN}$

(1)

(b) Use a vector method to find the ratio $AP:PN$

$$AP:PN = \dots\dots\dots$$

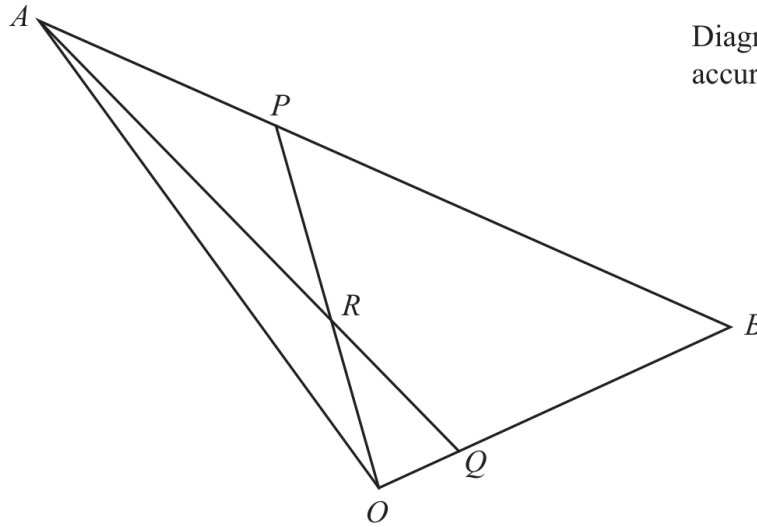
(4)

Question 16

4MA1-1H Q24 M01, M02, M03

MODULAR UNIT 2

6 marks

24 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

 ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

Question 16 (continued)

4MA1-1H Q24 M01, M02, M03

MODULAR UNIT 2

6 marks

(iii) $\vec{OR}$

(4)

Question 29

4MA1-2H Q17 M01

MODULAR UNIT 2

3 marks

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

.....

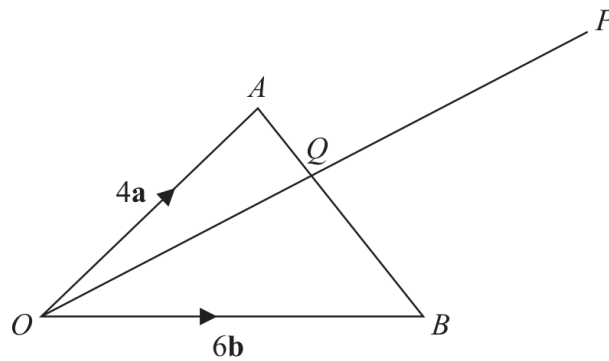
Question 30

4MA1-2H Q21 Q21

MODULAR UNIT 2

5 marks

21

Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ:QB$

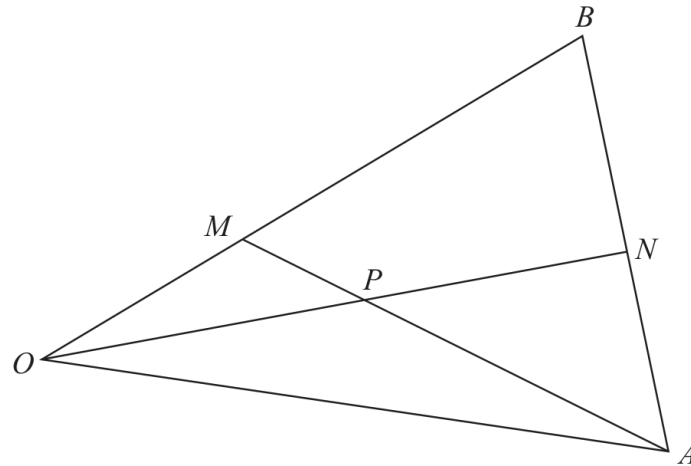
$$AQ:QB = \dots\dots\dots$$

Question 13

4MA1-1H Q22

MODULAR UNIT 2

5 marks

22 The diagram shows triangle OAB Diagram **NOT**
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

M is the point on OB such that $OM:MB = 1:2$

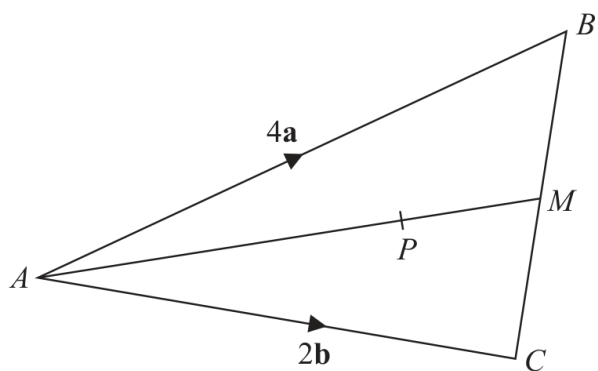
N is the midpoint of AB

P is the point of intersection of ON and AM

Using a vector method, find $\vec{OP}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

23

Diagram **NOT**
accurately drawn ABC is a triangle.The midpoint of BC is M . P is a point on AM .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio $AP:PM$

Diagram **NOT**
accurately drawn

$$\overrightarrow{OA} = 2\mathbf{a} \quad \text{and} \quad \overrightarrow{OB} = 2\mathbf{b}$$

N is the point on OB such that $ON:NB = 2:1$

P is the point on AN such that OPM is a straight line.

Use a vector method to find $OP:PM$
Show your working clearly.

A diagram of a parallelogram $OACB$ with origin O . The vertices are O (bottom-left), A (top-left), C (top-right), and B (bottom-right). The diagonals OC and AB intersect at point P . Vectors are defined as follows: $\vec{OA} = 2\mathbf{a}$, $\vec{OB} = 5\mathbf{b}$, and $\vec{AC} = 3\mathbf{b}$. The vector $\vec{AC}$ is shown as a horizontal arrow pointing from A to C .

Diagram **NOT**
accurately drawn

$$\overrightarrow{OA} = 2\mathbf{a} \quad \overrightarrow{OB} = 5\mathbf{b} \quad \overrightarrow{AC} = 3\mathbf{b}$$

Find and simplify an expression, in terms of **a** and **b**, for $\overrightarrow{OP}$
Show your working clearly.

$$\overrightarrow{OP} = \dots\dots\dots$$

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .

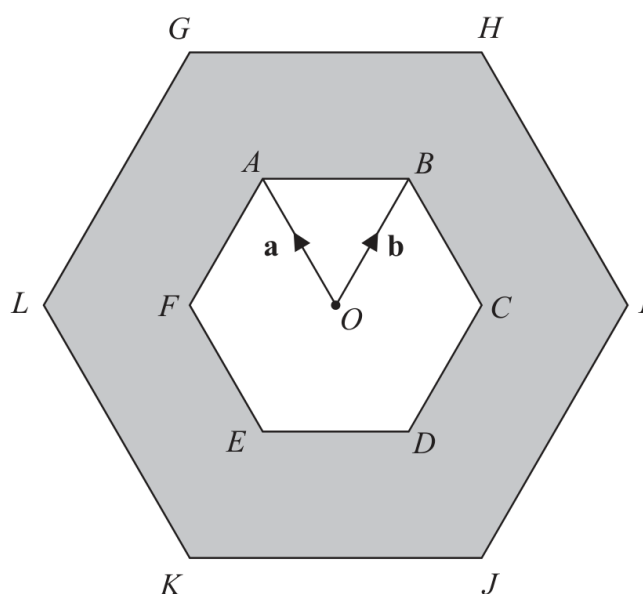


Diagram **NOT**
accurately drawn

$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

(1)

(ii) $\vec{KI}$

(2)

(iii) $\vec{LD}$

(2)

The triangle OAB has an area of 5 cm^2

(b) Calculate the area of the shaded region.

..... cm^2

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .

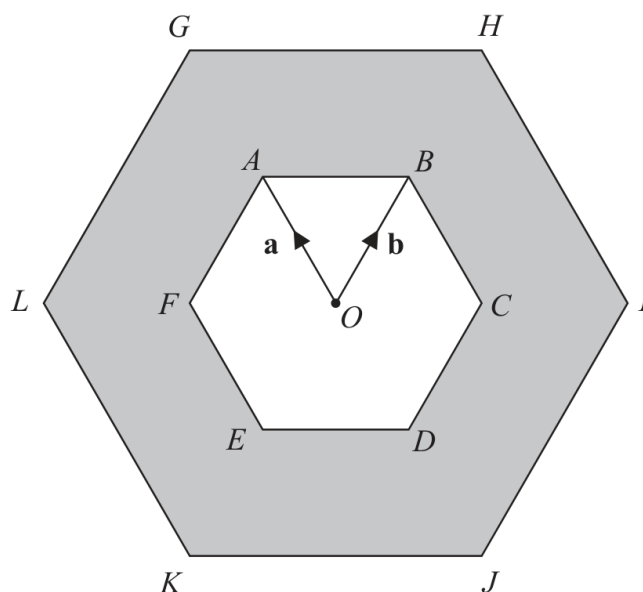


Diagram **NOT**
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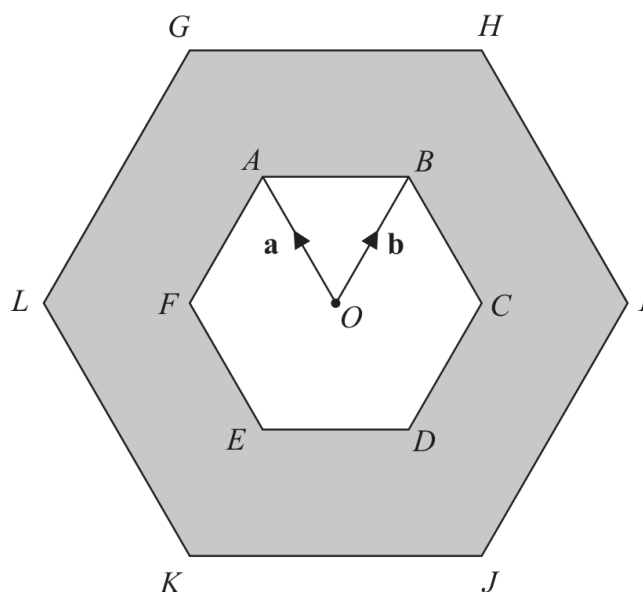


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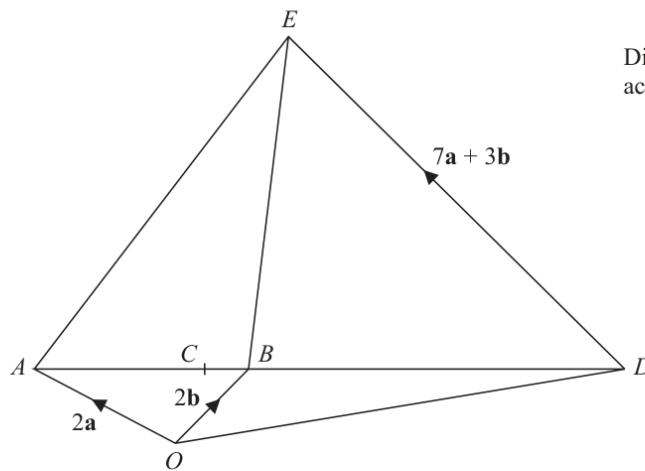
..... cm^2

$$\overrightarrow{OA} = \mathbf{a} \quad \overrightarrow{OB} = \mathbf{b}$$

The point D lies on OB such that $OD : DB = 1 : 2$

Using a vector method, prove that $ABDC$ is a trapezium.

Diagram **NOT**
accurately drawn



$$\overrightarrow{OA} = 2\mathbf{a} \quad \overrightarrow{OB} = 2\mathbf{b} \quad \overrightarrow{DE} = 7\mathbf{a} + 3\mathbf{b}$$

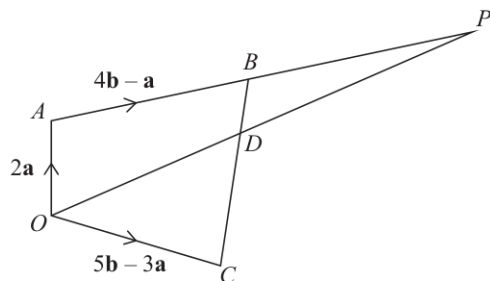
$$AB:BD = 1:2$$

The point C on AB is such that OCE is a straight line.

Use a vector method to find the ratio of $OC:CE$

(Total for Question 24 is 5 marks)

27

Diagram **NOT**
accurately drawn $OABC$ is a quadrilateral. ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$.
Simplify your answer.

(2)

The point D lies on BC such that $BD:DC = 1:3$ Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n

(Total for Question 27 is 6 marks)